Sanny Hong

AI Engineer · LLM Agents & Agentic Systems · Evaluation · Computer Vision · Time Series

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INTRODUCTION

AI engineer who builds LLM agents and machine-learning systems that run in production. In the technical department of an industrial-IoT company, I build agents that answer operators' questions over large operational databases, and computer-vision and time-series models that run on CPU-only hardware. I put the most care into knowing when a system is right: evaluation against ground truth, cost control and clear traces.

CORE SKILLS

LLM & Agents: tool calling, agent loops, context engineering, RAG and grounding, structured outputs
Evaluation & Reliability: ground-truth evaluation, deterministic tests, error analysis, cost and rate-limit control, run tracing
Machine Learning: PyTorch, OpenCV, segmentation, OCR, detection and tracking, anomaly detection, time-series forecasting
Data & Engineering: Python, SQL, TypeScript, FastAPI, Django, React, PostgreSQL, SQL Server, Docker, ETL, CI/CD

EXPERIENCE

Software Engineer — Flotech Controls, Singapore *Dec 2024 – Present*

Regional process-control and instrumentation provider. Technical department building AI systems for public water-infrastructure clients.

End-to-end ownership: from data preparation and model development to deployment, scheduling, alerting and monitoring in live industrial-IoT systems, with documentation for each system.

- **LLM agents:** Built and shipped tool-calling LLM agents that answer operators' questions over a 19M-row operational database through read-only, parameterised SQL tools, with a provider-agnostic model layer, so models can be swapped without changing agent code.
- **Cost control:** Cut tokens per question by 84% through context engineering and tool gating, and enforced per-request spending caps in production.
- **Evaluation:** Built ground-truth evaluation against live data, which caught a silent join error no user had reported; fixed it with schema-relationship context and added entity resolution for location names.
- **Computer vision:** Developed a flood-level reading pipeline that fuses a fine-tuned segmentation model, several detectors and OCR calibration, bringing reading error down to about one pixel.
- **Model re-engineering:** Rebuilt a GPU vision pipeline as a CPU-only OpenCV stack with the same accuracy on the same test set, and added automatic labelling for retraining.
- **Anomaly detection:** Built anomaly detection for high-frequency water-quality data with ETL, scheduling and alerting, and moved the storage to a star schema for faster analytics.

Independent Projects & Further Study — Singapore *Dec 2022 – Nov 2024*

Software Engineer (Frontend Lead) — GienTech, Shanghai *Feb 2022 – Dec 2022*

Seconded to OCBC. China-based expert in digital-transformation strategy and solutions.

- Led front-end development of OCBC Malaysia's financial-record system for 1,000+ users across its full lifecycle, and mentored junior engineers.
- Delivered an HR management system in React, TypeScript and Django, and cut the codebase by about 40% during a TypeScript migration by consolidating shared logic.

PERSONAL PROJECTS

TapeSense — Macroeconomic Knowledge Graph with AI Agents · tapesense--preview.expo.app *2026 – Present*

- Applied knowledge-graph modelling to macroeconomics as a cause-and-effect graph, with an evidence pipeline that checks each chain's predicted market moves against real prices after each data release.
- **In progress:** a self-updating graph where each link learns how often it holds, and LLM agents that use it as read-only tools to explain markets and draft updates for human review.

Agentic AI — Forecasting Engine *2023 – Present*

- Natural-language agent over a recommender and demand forecaster that answers only from real catalogue items and declines when nothing matches. github.com/HSanny/forecasting-engine

Computer Vision — Motion Analyser *2026 – Present*

- Turns a single camera feed into line-crossing, dwell-time and speed events with detection and multi-object tracking. Real-footage trials in progress. github.com/HSanny/motion-analyser

EDUCATION

Master of IT in Business (Artificial Intelligence) — Singapore Management University · Scholarship *2024 – Present (On Leave)*

Bachelor of Information Technology — The Australian National University *2017 – 2022*

Majors: Computer Science & Mathematics · Specialisation: Intelligent Systems · Minor: Japanese Language & Culture

LANGUAGE

Languages: English (full professional) · Mandarin, Cantonese (native) · Japanese (conversational)